Chest X-Ray Pneumonia Classification

A Comparative Study of Logistic Regression, CNNs, ViTs, and Hybrid Architecture

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- Amrithavarshini Satheesh — CNN Baseline
- DivyaTej Reddy Kavalagari — Vision Transformer Baseline
- Sohith Vishnu Sai Yachamaneni— Proposed Hybrid ViT Model

Why This Problem Matters

- Pneumonia is a major respiratory disease
- Chest X-rays are commonly used for diagnosis
- AI can support radiologists by improving consistency and speed

Formal Problem Setting

Input: Chest X-ray image $x_i \in \mathbb{R}$^(H×W×C)

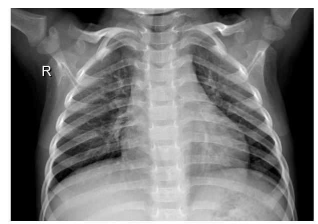
Output label $y_i \in \{0,1,2\}$

- 0 = Normal (i.e. no Pneumonia)
- 1 = Bacterial Pneumonia
- 2 = Viral Pneumonia

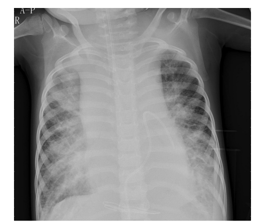
Supervised three-class image classification problem

Dataset Introduction

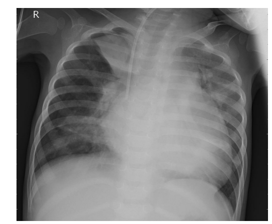
- Chest X-Ray Pneumonia Dataset*
- Classes:
  - Normal: Clear lung fields
  - Bacterial: Typically focal lung opacity
  - Viral: Diffuse and widespread involvement



Normal



Viral



Bacterial

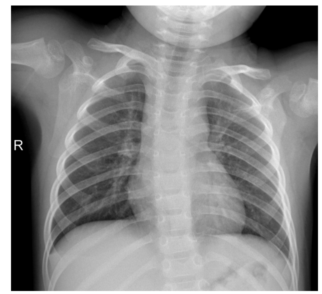
https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

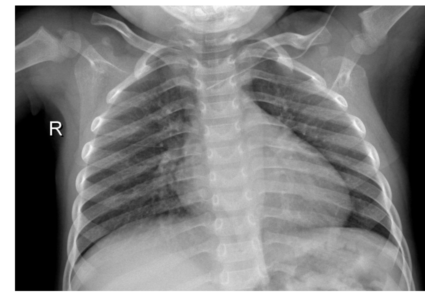
Dataset

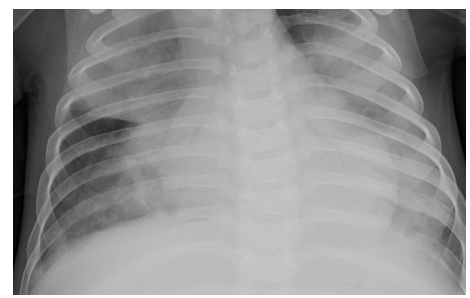
- Approximately 5,800 chest X-ray images

Images contain:

- Different patient positioning
- Varying image quality
- Machine-specific artifacts
- Different coverage regions







https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

Dataset Distribution

- Significant class imbalance exists
- Validation set is extremely small
- Bacterial pneumonia is the dominant class

Split	Total	Bacterial	Viral	Normal
train	5216	2530	1345	1341
val	16	8	-	8
test	624	242	148	234

Dataset Analysis

- All images are converted to a consistent 3-channel format during preprocessing
- Data is pre-cleaned up. However pixel intensity statistics are analyzed for outliers using:
  - Min / max intensity
  - Mean brightness
  - Contrast
- Observed intensity differences are likely caused by scanner or acquisition variability

Preprocessing & Augmentation

- All images are downscaled to 224x224
- Augmentation during training
  - Random rotation up to ±10°
  - Translation up to 3 pixels
  - Gaussian noise with standard deviation 0.02
  - Brightness adjustment up to ±0.10
  - Augmentation probability: 0.5

Evaluation Metrics

- Macro Recall is chosen as:
    - Robust to class imbalance
    - Gives equal importance to all classes
    - Measures sensitivity across classes

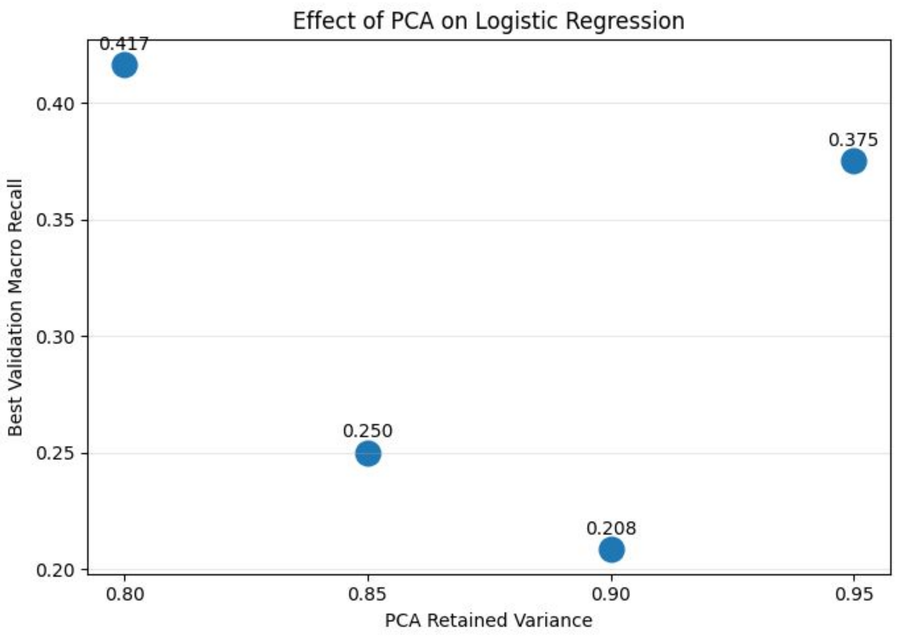
Baseline 1 — Logistic Regression

Logistic Regression as an ML baseline:

- Flatten each image into a one-dimensional feature vector
- Z-score standardization across training set
- PCA for dimensionality reduction
- Class-balanced weights to address dataset imbalance

Baseline 1 — Logistic Regression

- Best PCA: 0.80
- Best validation Macro Recall: 0.417
- Regularization strength had minimal impact on performance

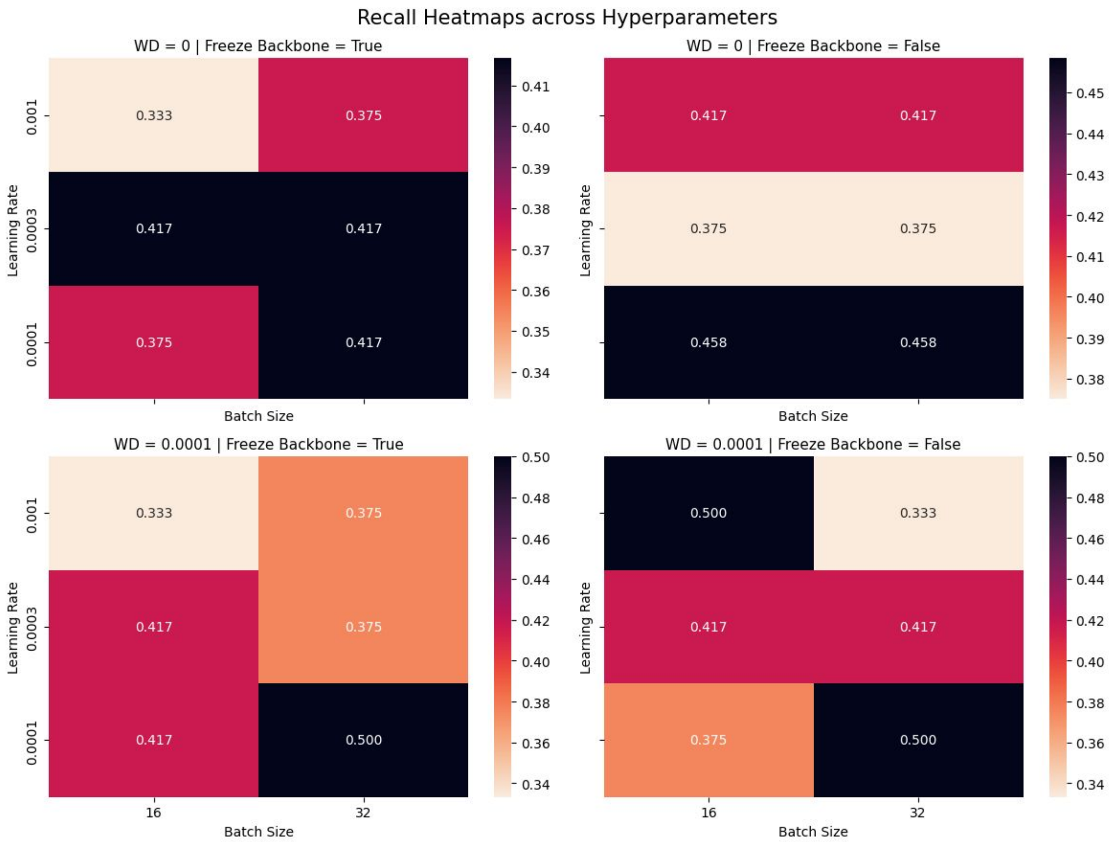


Baseline 2 — CNN Model

- DenseNet-121 architecture
- Through convolutions, the model is able to process local image regions
- In medical imaging, pneumonia appears as localised texture patterns and opacities, and so CNN would be a good choice

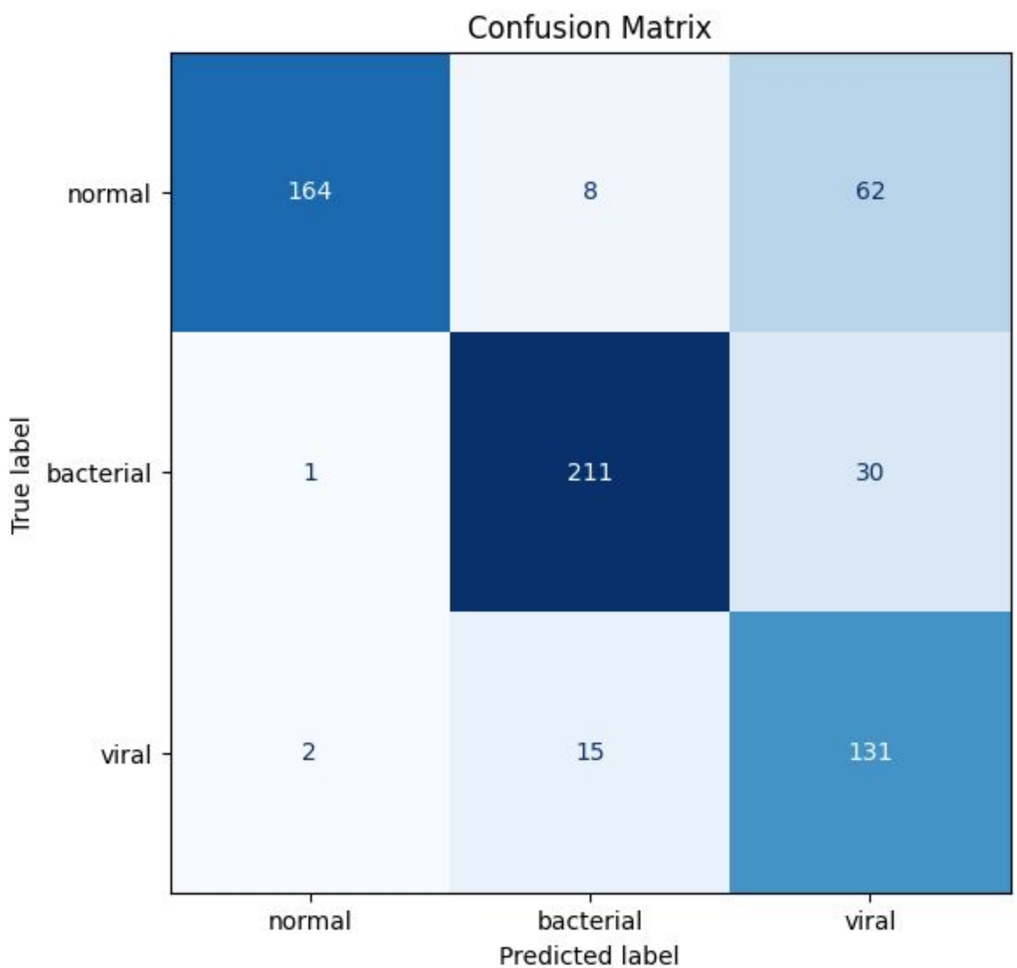
Baseline 2 — CNN Model

- Through hyperparameter tuning, we set
  - Learning rate = 0.001
  - Weight Decay = 0.0001
  - Batch Size = 16
  - Backbone is not frozen, so the entire network is fine-tuned



Baseline 2 — CNN Model

- Model is trained on augmented data
- We use Cross Entropy Loss as a loss function so that minority-class mistakes are penalised more

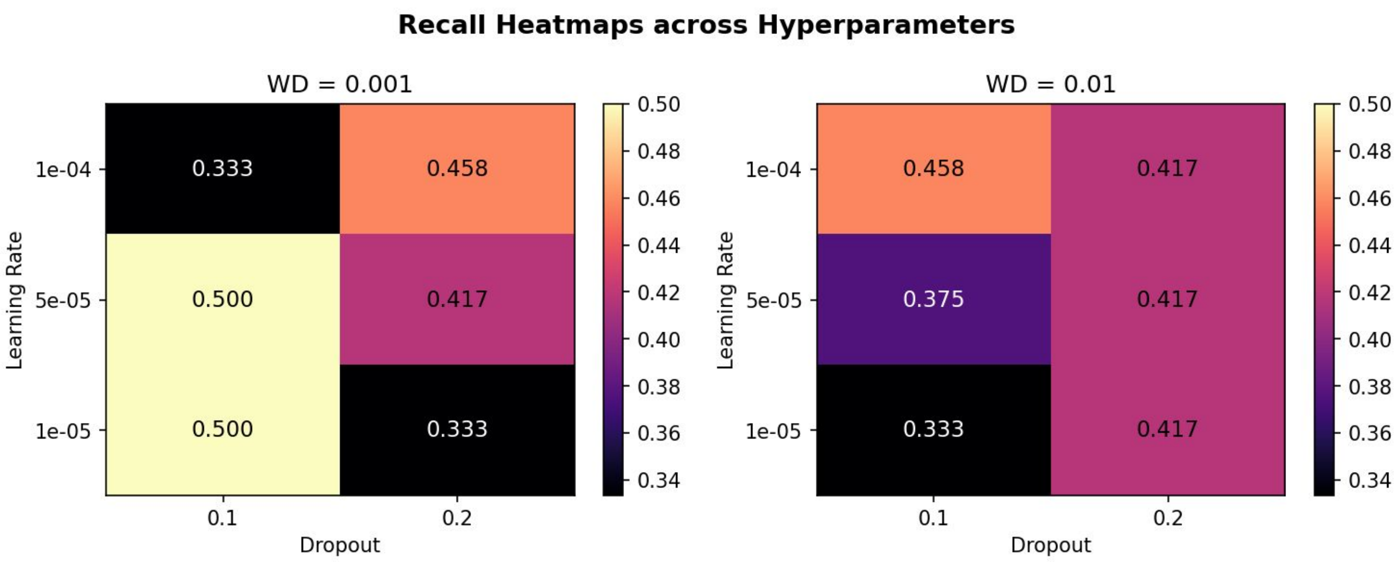


Baseline 3 — Vision Transformer (ViT)

- ViT processes images as sequences of patches
- Pipeline:
  - Patch extraction
  - Patch embeddings
  - Transformer encoder
  - Classification using CLS token
- Strength: Global context modeling

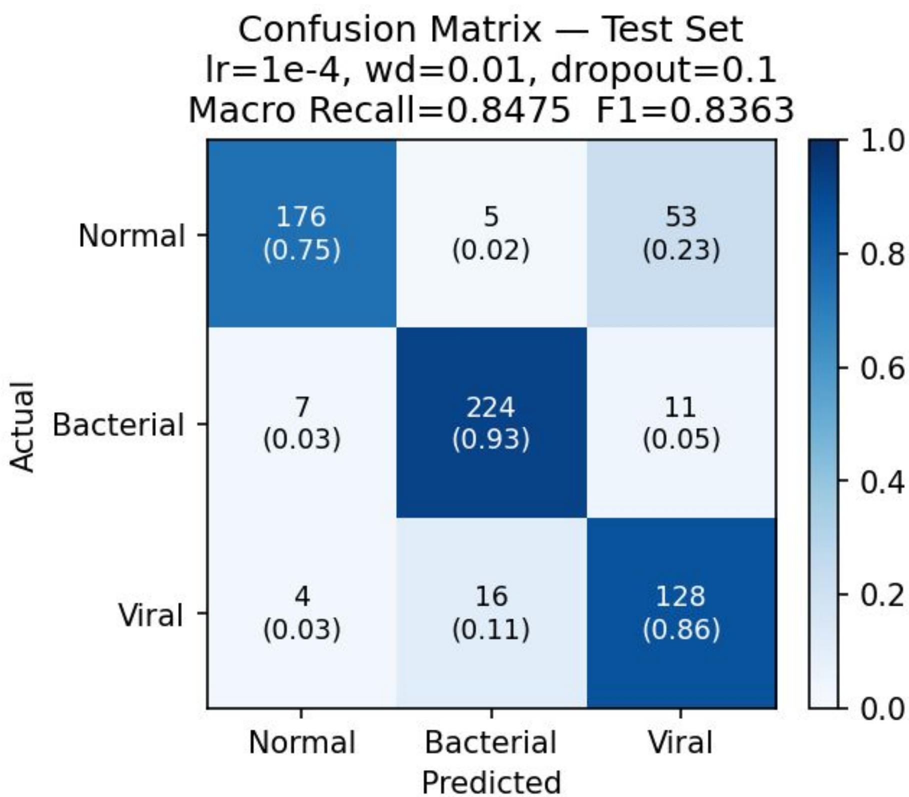
ViT Hyperparameter Tuning

- Through hyperparameter tuning, we set
  - Learning rate = 1e-4
  - Weight Decay = 0.01
  - Dropout = 0.1
  - Backbone is not frozen - so the entire network is fine-tuned



ViT Results

- Model is trained on augmented data
- Weighted Cross-Entropy Loss with label smoothing to handle class imbalance



Challenges with Standard ViTs

- Problems on small datasets:
  - Weak spatial inductive bias
  - Data hungry
  - Overfitting risk

- Observation:
  - CNNs = better local features
  - ViTs = better global structure

Proposed Hybrid Architecture

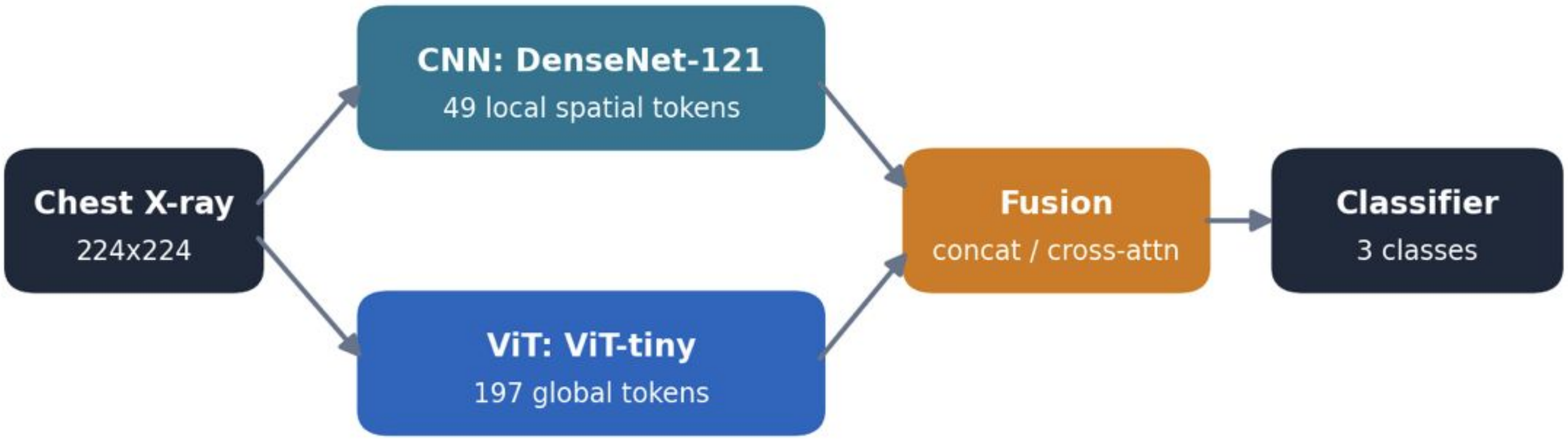
Two parallel branches, each kept at full token resolution

CNN — DenseNet-121: keeps 49 local spatial tokens (texture, focal opacities).

ViT — ViT-tiny: 197 global patch tokens for overall lung structure.

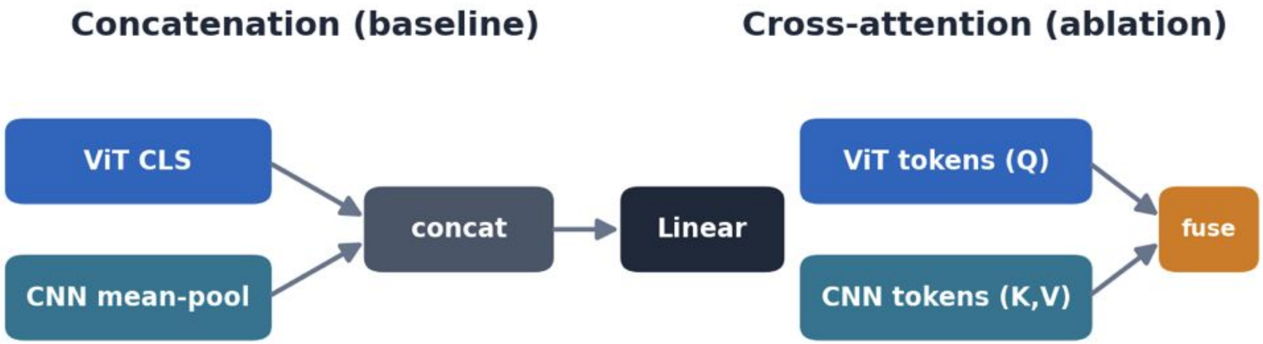
Fusion → classifier: the two token sets are combined, then classified into Normal / Bacterial / Viral.

Both branches keep their spatial tokens (not collapsed to a single vector), so fusion can relate local detail to global context.



Fusion Mechanism & Regularization

- **Concatenation (baseline)**: ViT CLS token + mean-pooled CNN tokens → a linear classifier.
- **Cross-attention (ablation)**: ViT patch tokens query the CNN spatial tokens via multi-head attention, so global features pull in the most relevant local detail.
- **Regularization**: embedding-level noise / dropout + class weighting to counter imbalance and overfitting.



Fusion Mechanism

Concatenation

- ViT CLS token + mean-pooled cnn map gives one vector and linear classifier.
- Fixed combination, the vit and cnn branch doesn't interact before classifier

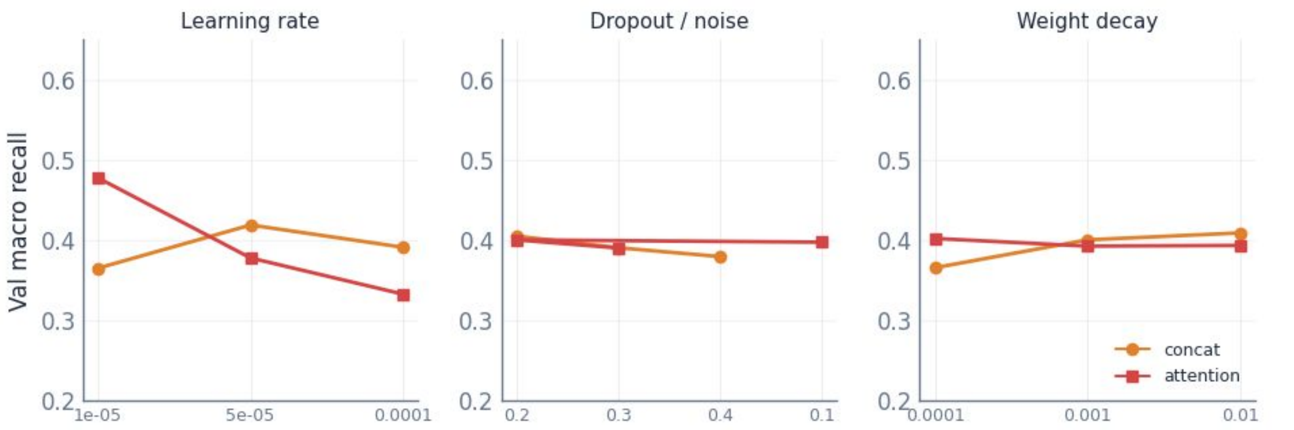
Cross-Attention

- ViT tokens query the CNN spatial tokens (multi-head), then pool
- Learned interaction, but a fresh attention block to train, which is more Hyperparameter sensitive harder to optimize

Fusion	Test Recall	Normal Recall	Bacterial Re	Viral Recall
concat	0.7409	0.5342	0.9587	0.7297
attention	0.6949	0.8547	0.7975	0.4324

In this experiment setup concatenation has performed better as cross attention's extra block was harder to train and collapsed on viral, the hardest and most fusion depend class - so concatenation is our hybrid setup here in the final results.

Hyperparameter Tuning



Learning rate dominates — dropout and weight decay barely move validation recall.

Two-phase search, run separately for each fusion type

27 configs (learning rate × dropout × weight decay) screened for 8 epochs → best configs retrained for 20 epochs.

Selected on a stratified fold held out from train — the official 16-image val (with 0 viral cases) is too small to select on.

Best config	LR	Dropout	Wt decay
concat	5E-05	0.2	1e-4
attention	1E-05	0.2	1e-4

Results

	LR	CNN	ViT	Hybrid (Concat)
Macro Recall	0.69	0.82	0.84	0.74

Comparison

- Logistic Regression: Simple but limited
- CNN: Strong local feature extraction
- ViT: Strong global context
- Hybrid Model: Combines local and global representations

Model	Macro Recall	Normal Recall	Bacterial Recall	Viral Recall
Logistic Regression	0.69	0.43	0.89	0.75
CNN(DenseNet 121)	0.82	0.70	0.88	0.88
ViT	0.84	0.75	0.93	0.86
Hybrid (Concat)	0.74	0.53	0.95	0.72

Conclusion

- The standard ViT baseline model performed better than our proposed hybrid architecture
  - A significant limitation is the dataset imbalance,
  - The validation set contains only 16 images, with no viral pneumonia cases, which limits its ability to provide a reliable estimate of model performance during hyperparameter tuning and model selection
  - The hybrid -complex- architecture is not able to generalise well, especially given the small and imbalanced dataset
- Future work could explore with better split of dataset to improve diagnostic accuracy